
Education

University of California, Davis**Expected December 2026**

- Bachelor of Science in Electrical and Computer Engineering
- IEEE Member, AvenueE Scholar, Trio SSS Member, EOP Member, SHPE Member
- GPA: 3.67, Bilingual (English/Spanish)

Skills

Programming: Python (NumPy, SciPy, Matplotlib), MATLAB, C, C++, Verilog, x86 Assembly, Git, RISC-V, Linux, HTML**Hardware & Optics:** Circuit design, optical test setups, free-space optics, laser systems, optical tables, fiber coupling, photodetectors, spectrometers, oscilloscopes, function generators, power supplies, CMOS/BJT devices, SEM microscopy**Microfabrication & Process Engineering:** Photolithography, dry/wet etching, PECVD/PVD, wafer cleaning, thin-film deposition, silicon photonics device fabrication, process monitoring, layout-to-fabrication validation, 2D materials

Internship Experience

Photonics Research Intern (Ekinci Lab), Boston University — Boston, Massachusetts, US**June 2025 — August 2025**

- Designed and optimized cleanroom microfabrication process flows for silicon photonics-compatible NEMS devices
- Applied machine learning and signal processing to characterize semiconductor device behavior and performance yield
- Developed Python-based test and data acquisition frameworks to automate nanoscale measurements and analysis

Machine Learning Research Intern (CGM Center), UC Davis — Davis, California, US**April 2025 — August 2025**

- Created a signal and image analysis pipeline for automated feature detection in high-resolution experimental datasets
- Applied frequency-domain and image processing methods to improve feature localization and classification accuracy
- Optimized workflows for datasets, supporting automated inspection and yield analysis using convolutional algorithms

Photonics Research Intern (Moody Lab), UC Santa Barbara — Santa Barbara, California, US**June 2024 — August 2024**

- Performed cleanroom fabrication and photoluminescence characterization of 2D integrated photonic devices
- Executed precision optical alignment of lasers and spectrometers to maximize laboratory photon collection efficiency
- Designed photonic chip layouts and components using KLayout and SolidWorks for nanofabrication and device testing

Relevant Coursework

Semiconductor and Device Physics I, Electromagnetics I, MOS Circuit Design, BJT Transistor Design, Analog Signals & Systems I, II, Digital Signals & Systems I, II, Algorithm Design & Analysis I, II, Microcontrollers & Embedded Systems, Digital Design Capstone, Computer Architecture, Operating Systems, Computer Networks

Projects

Optical Photoluminescence Characterization System

- Designed with Ansys and assembled an optical test system with lasers, mirrors, filters, and a spectrometer with Thorlabs equipment for 2D hBN characterization, improving material data collection by 12%
- Improved photon collection efficiency by 7% using precision mirror alignment, beam focusing, and optical alignment
- Created CAD and 3D Blender models to document the setup, visualize optics, and support device validation

CMOS-Compatible Silicon Device Fabrication

- Executed silicon wafer fabrication flows using photolithography, wet/dry etching, and Class 100 cleanroom processes
- Performed critical process steps, including substrate cleaning, thermal oxidation, thin-film deposition (PECVD/sputtering), and mask alignment for high-precision pattern transfer and doping on single-crystal Si wafers
- Designed and fabricated a NEMS resonator prototype using CMOS-compatible processes and laser interferometry

Digital Dice Game 23

- Designed and implemented a digital control and test sequencing system in Quartus using schematic design and HDL
- Developed modular FSM-based control logic, including counters and correction logic, to manage system states, test flow, and real-time validation sequencing using Verilog
- Integrated LED and seven-segment outputs to verify real-time system behavior, timing, and functional correctness

Awards, Certifications & Leadership

- President of the SHPE chapter at Oxnard College, SHPE Northrop Grumman scholarship recipient
- Altium Designer Certification (PBC/chip design), ITTPC Certification (200+ hours in math/physics/chemistry tutoring)